

GalaxySagittarius-AF: Predicting Targets for Drug-Like Compounds in the Extended Human 3D Proteome

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Abstract

In recent years, advancements in deep learning techniques have significantly expanded the structural coverage of the human proteome. GalaxySagittarius-AF is the first method that translates these achievements in structure prediction into target prediction for druglike compounds by incorporating predicted structures. This web server searches the database of human protein structures using both similarity- and structure-based approaches, suggesting potential targets for a given druglike compound. In comparison to its predecessor, GalaxySagittarius, GalaxySagittarius-AF utilizes an enlarged structure database, incorporating curated AlphaFold model structures alongside their binding sites and ligands, predicted using an updated version of GalaxySite. GalaxySagittarius-AF covers the largest human protein space among available computational target screening methods. Its structure-based prediction method maximizes the utility of the expanded structural information, setting it apart from other target prediction servers that rely on ligand-based prediction methods. Additionally, the web server has undergone enhancements, operating two to three times faster than its predecessor. The updated report page provides comprehensive information on the sequence and structure of the predicted protein targets. GalaxySagittarius-AF is accessible at http://galaxy.seoklab.org/sagittarius_af without the need for registration.

1. Introduction

The introduction of innovative artificial intelligence techniques for predicting protein structures from sequences^{1,2} has resulted in the expansion of the structure database. It is essential to disseminate this extended structure database across various biopharmaceutical research domains. One direction involves assessing the impact of drug-like molecules on the human biological system based on predicted structures, using cost-effective *in silico* methods.

As a first step in this direction, we have developed a free web service to identify protein targets with which a given drug-like molecule can interact within the expanded human 3D proteome. This proteome comprises both experimentally resolved structures and those predicted by AlphaFold. The web server not only provides information on the potential interactors' identities but also offers specific three-dimensional interaction models.

Compared to other target prediction methods³⁻⁵ that heavily depend on available activity data, GalaxySagittarius⁶ remains the only available target prediction webserver employing structure-based and docking-based methods for target prediction. As described in detail in our previous paper⁶, GalaxySagittarius takes the query compound as input, searches against the database of human protein structures, and returns potential protein targets for the compound. During the search against the target proteins, it utilizes both ligand similarity-based and protein structure-based approaches as efficient screening methods, followed by protein-compound docking for rescoring. This process returns predicted protein targets along with their respective protein-compound complex structures.

One of the crucial components of GalaxySagittarius revolves around the protein target database used in the target prediction process. Given that the prediction method includes structure-based methods, atomic protein structures and their binding sites are required. Additionally, for the ligand similarity-based search, comprehensive binding ligand information is required. Thus, it is important that our database is equipped with a large and high-quality collection of protein structures and interaction information covering diverse human proteins.

Compared to GalaxySagittarius⁶, GalaxySagittarius-AF utilizes an updated protein structure database that integrates recently resolved experimental structures from the PDB database⁷, and the predicted structures from the AlphaFold Protein Structure Database (AFDB)⁸. For both experimental and predicted structures, the binding sites and ligands were newly predicted using an updated version of GalaxySite, leading to broader coverage of targets with binding information and a wider range of ligand varieties. The updates greatly expanded the protein target database and improved the target prediction accuracy of GalaxySagittarius-AF.

2. Updates in the protein target database

The protein target database used in both GalaxySagittarius and GalaxySagittarius-AF comprises of three main components: protein structures, their binding sites, and the corresponding binding ligands. In the earlier version, GalaxySagittarius⁶, the experimentally resolved human protein structures were gathered to generate the structure database. The binding site and ligand information were initially extracted from the experimental protein-ligand complex structures and further predicted using template-based method GalaxySite. Compared to GalaxySagittarius, GalaxySagittarius-AF introduces two major updates: the expansion of the protein structure repository, incorporating recent experimental structures and the model structures predicted using AlphaFold2⁸, and the use of the updated GalaxySite^{9, 10} for binding site and binding ligand prediction for both experimental and model structures.

Among the predicted protein structures for 48 organisms offered by AFDB¹¹, the structures encompassing the human proteome was used to augment the structure database for GalaxySagittarius-AF. These structures were curated in advance to include only the high-quality structure domains. Given that the predicted structures can include poorly modeled regions that might disrupt the target prediction process, efforts were made to remove the low-quality regions and splice the full-length structures into adequate structural domains. Further details and insights into the structure curation process are available in the corresponding paper¹⁰.

The curated structures sourced from AFDB were then merged with the experimental human protein structures including PDB⁷ entries up to 2023-08-15. The structures were subsequently processed to extract the experimentally resolved binding site and binding ligand information, followed by a preparation step to ready them for future docking-based screening. The extracted ligands encompass non-polymeric entries, excluding covalently bonded ligands and cofactors, while cofactors were tracked for use in the docking step. For each experimental and model structure, the updated GalaxySite^{9, 10} was further employed to predict potential binding sites and ligands. The key enhancement in the updated GalaxySite involves the use of an updated ligand interaction database during the search, resulting in improved binding site prediction performance¹⁰ and the augmented number of predicted sites and ligands, as discussed in section 2.1. P2Rank¹² was utilized to predict binding sites for the structures with no predicted sites by GalaxySite. The information on the experimental and predicted binding sites and ligands underwent initial clustering within each structure, followed by re-clustering within the proteins with the same UNIPROT IDs, using a hierarchical clustering algorithm with a distance cutoff of 10 Å for each structure.

2.1. Database statistics

The integration of predicted structures significantly expanded the protein target database. GalaxySagittarius-AF now searches against 71,425 human protein structures, covering 15,956 unique proteins. This number surpasses the 6,037 unique protein targets covered by GalaxySagittarius, as well as other target prediction webserver such as SwissTargetPrediction³ (3,068) and SuperPred 3.0⁴ (2,353). The primary reason for the expanded coverage is the incorporation of predicted structures. Notably, among the 2,994 human protein targets with available activity data in the ChEMBL database (ChEMBL33), the updated database used in GalaxySagittarius-AF covers 2,939 targets, a substantial improvement over the previous version, which covered only 1,960 targets.

GalaxySagittarius-AF runs in one of the two modes: compatibility mode and combined mode, just like the previous version. The compatibility mode fast-evaluates the structural compatibility of the given ligand with protein structures in the database before docking. The other available mode is the combined mode, which performs an additional ligand similarity search. This mode requires at least one known/predicted ligand for each protein, so the number of proteins searched in this mode is smaller, totaling 15,518 unique proteins. The number of proteins, binding sites, and ligands in the target database is presented in **Table 1** for each mode. The number of binding sites and ligand interactions per target structure in the database increased 2-fold and 12-fold, respectively, from the previous version.

Table1. The size of each entry in the protein target database of GalaxySagittarius and GalaxySagittarius-AF

Type of entry	GalaxySagittarius		GalaxySagittarius-AF			
	2019 PDB		2023 PDB		2023 PDB + AF	
	Compatible mode	Combined mode	Compatible mode	Combined mode	Compatible mode	Combined mode
# protein sequences	6,037	4,454	7,479	6,694	15,956	15,518
# protein structures	41,333	33,381	56,791	54,298	71,425	68,932
# binding sites per structure	3.24	3.39	9.04	9.15	8.15	8.20
# unique ligands	13,637		24,771		26,875	
# ligands per structure	6.97		99.16		84.11	

3. Benchmark Test of Protein Target Search

3.1. Benchmark methods

Construction of a benchmark compound set: To assess the performance of the server, a set of 543 non-redundant compounds was compiled from the most recent version (ChEMBL33) of the ChEMBL¹³ database, using the same curation criteria as before⁶. Specifically, from the newly included activity data in ChEMBL33 from the 32nd version, the ones with high activity M (Ki, Kd, IC50, or EC50) with pM ≥ 5.5 were selected. Compounds with at least one target present in our database were subsequently clustered using a Tanimoto coefficient cutoff of 0.5 to select non-redundant compounds. The benchmark set has an average of 5.73 known, considered as ground truth, human protein targets per compound in ChEMBL33.

Compared methods: For the benchmark, we utilized the GalaxySagittarius pipeline (“2019 PDB”) and the updated pipeline, with and without AlphaFold2 models included in the database (“2023 PDB + AF2” and “2023 PDB”, respectively). Each pipeline operates in two distinct modes: a structure-based compatibility mode (“compatibility mode”) and a combined ligand similarity and structure-based mode (“combined mode”). The pre-docking screening in the compatibility mode predicts the match between the binding sites in each protein structure and the input query ligand. This structure-based search covers a broader range of protein targets, without the need for prior interaction data. The pre-docking screening of the combined mode integrates an additional similarity search to the known/predicted ligands of the database proteins. The top 1,000 candidate proteins selected using the pre-docking screening of each mode are then subjected to protein-ligand docking using GalaxyDock2¹⁴ for final ranking through atomic level analysis.

3.2. Benchmark results

The target prediction performance on the benchmark set was assessed via two metrics: First, the average success rate per each compound, where successful prediction is defined as at least one target found in the top N predicted targets [Figures 1 (a) and 1 (b)], where N = 100, 200, and 500. Second, the average percentage of known targets recovered among the top N predicted targets, to assess the ability to recover multiple known targets [Figures 1 (c) and 1 (d)]. The numerical data used for the figures are provided in Supplementary Table S1. Results for a protocol without using the time-consuming docking step are also presented in Supplementary Table S2.

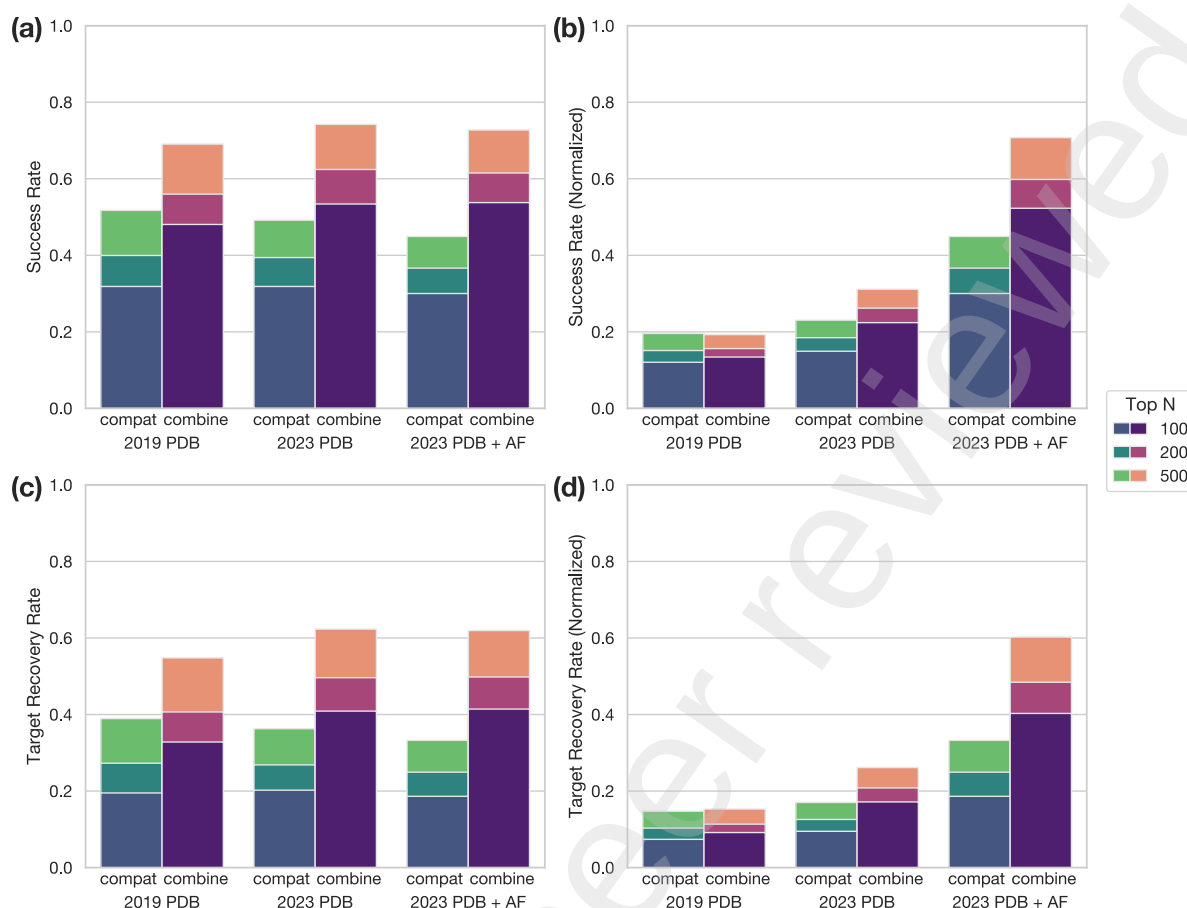


Figure 1. Success rate and target recovery rate for different target prediction pipelines: “compat” denotes the compatibility mode, and “combine” denotes the combined mode. The success rates in (a) and the target recovery rates in (c) were normalized by multiplying them by the ratio of the number of proteins in each database relative to that of “2023 PDB + AF” in (b) and (d), respectively, to correct for the different database sizes.

The updated version demonstrated improvements in both success rate and target recovery rate. For instance, when using the combined mode with the full database including AlphaFold structures for target prediction, the recovery rate among the top 100 predicted targets increased from 32.8% with GalaxySagittarius (“2019 PDB”) to 41.4% (“2023 PDB+AF”), as shown in **Figure 1 (c)**. The actual enhancement of GalaxySagittarius-AF is much more significant, considering that the same number of candidates were selected from the much-enlarged database. The recovery rate normalized by the number of unique proteins within each database is illustrated in **Figure 1 (d)**. The substantial decrease in the normalized recovery rate for pipelines with smaller databases (“2019 PDB” and “2023 PDB”) illustrates the successful enrichment of experimentally revealed targets among the top predicted targets by GalaxySagittarius-AF (“2023 PDB+AF”).

The incorporation of AlphaFold structures allowed GalaxySagittarius-AF to detect potential targets even in the absence of experimentally resolved structures. Using the compatibility (and combined) mode on the full database, we were able to recover 22/32/72 (and 4/16/50) targets solely from predicted structures for 14/17/28 (and 4/ 8/15) benchmark compounds within the top 100/200/500 predicted targets. These figures demonstrate that GalaxySagittarius-AF possesses the capability to thoroughly explore and leverage the extended database. One of the successful examples is the target prediction result for a potential NIAK inhibitor¹⁵ (ChEMBL ID:

CHEMBL5199998) with six identified protein targets in the ChEMBL database, none of which currently have available experimentally determined structures. All six targets belong to the AMP-activated protein kinase subfamily, with high-quality model structures available. GalaxySagittarius-AF's combined mode was able to recover five of six targets in the top 100 and the last target as the 123rd. Analysis of the predicted binding poses revealed that GalaxySagittarius-AF predicted the known binding sites for all six kinases with high confidence, as shown in Supplementary **Figure S1**. This example serves as an illustration of how high-quality model structures elevate target prediction accuracy.

The combined mode, which utilizes the known/predicted binding ligands in the database, showed higher target search performance than the compatibility mode across all databases. The significant enhancement for the combined mode using recent databases ('2023 PDB' and '2023 PDB + AF' over '2019 PDB') is attributed to the update in the binding site and ligand prediction method, GalaxySite. This update enabled a similarity-based search against proteins that weren't covered in the earlier version. The new version also provides enhanced ligand interaction information with a doubled number of unique ligands (**Table 1**).

Although utilizing available information on ligand interaction resulted in higher performance for the combined mode, the compatibility mode still has the advantage when information on binding partners is unavailable. One such example is the case of a potential Pyruvate Kinase M2 (PKM2) inhibitor¹⁶ (ChEMBLID: CHEMBL5189951). The sole known target of the inhibitor, PKM2 (UniProtID: P14618), ranked 103rd in the compatibility mode, while the same protein failed to advance beyond the pre-screening step in the combined mode. The target, P14618, had 48 known and predicted binding ligands in the database. Among them, the most similar ligand to the inhibitor had the Tanimoto coefficient of only 0.16, resulting in a lower rank compared to other targets with more closely related binder information available. This example demonstrates how the compatibility mode, focusing on structural details, can compensate for the combined mode, particularly for novel compounds and targets.

4. Webserver interface

GalaxySagittarius-AF operates as a standalone web server, featuring upgraded functionalities in both its backend and frontend compared to its predecessor, GalaxySagittarius. GalaxySagittarius-AF now operates at a speed two to three times faster even with the much larger database. This boost is achieved through a combination of an optimized search process and updates in the protein-ligand docking protocol of GalaxyDock2¹⁴. It takes an average of 5 minutes without docking and 2.5 hours with docking, which is a notable improvement from the previous version which took 10 minutes without docking and 6 hours with docking.

Figure 2 displays an example of the submission and the report page on the GalaxySagittarius-AF web server. The submission page offers users assistance and examples as guidance. The input ligand is accepted in various formats: a linear notation SMILES or a 3-dimensional structure in various formats. The webserver allows users to choose from a range of search methods: the two search modes (compatibility mode and combined mode), selection of structure database (with or without AlphaFold structures), and optional re-ranking using docking.

The report page has also been updated from the previous version to provide a more comprehensive details regarding the top predicted human protein targets. Users can access direct links to both structure (RCSB PDB⁷) and sequence (UniProt¹⁷) information for the predicted targets, facilitating direct navigation to the associated pages. Furthermore, for cases where an AlphaFold model structure is used to detect the potential target, users can

now seamlessly access the related HProteome-Bsite¹⁰ page, which offers detailed binding site information on the target.

Webserver submission page

GalaxyWEB
A web server for protein structure prediction, refinement, and related methods
Galax Inc. & Computational Biology Lab, Department of Chemistry, Seoul National University

Home Services Database Design Queue Help Softwares Suppl

GalaxySagittarius-AF

Prediction of potential targets of drug-like compounds using AlphaFold models

User Information

Job name

E-mail address (Optional)

Input ligand

Ligand File (≤150 atoms)

Or SMILES string (≤150 atoms)

Search Mode

☐ Similarity combined prediction

☐ Binding compatibility prediction

Structure database

☐ PDB + AlphaFold models

☐ PDB only

Re-ranking using docking

☐ yes

☐ no

Submit

submit reset

Help

Information

- Ligand File: It should be in mol2 (highly recommended), pdb, xyz format or smi(SMILES). For file formats other than mol2, OpenBabel will be used to convert the file format.
- E-mail: Average run time ranges from 5 minutes (without docking) to 2.5 hours (with docking). If e-mail address is given, the server sends notifications automatically. If not, the user has to bookmark the report page.

Example

- Ligand file: input_ligand.mol2
- Report: [View]

Information and examples related to the webserver

Inputs related to the query ligand and target search methods

HProteome-BSite
An online database for ligand binding sites in human proteomes
Computational Biology Lab, Department of Chemistry, Seoul National University

Search Help Entries Downloads

Get Your BSite

Query domain: P28838_F1_d1

Show Structure on PTV viewer [View]

- Download structural domain in PDB format
- See structural domain only (PTV viewer)

Webserver report page

Predicted binding poses

Predicted binding pose for top10 predicted targets

View in Mol* [Model 1] [Model 2] [Model 3] [Model 4] [Model 5] [Model 6] [Model 7] [Model 8] [Model 9] [Model 10]

Download [Rank 1] [Rank 2] [Rank 3] [Rank 4] [Rank 5] [Rank 6] [Rank 7] [Rank 8] [Rank 9] [Rank 10]

Predicted Target Information

Rank	PDB ID	Uniprot ID	Predock score	Docking score	Score
1	5n6_A	P09960	0.922	-21.247	1.134
2	4xx8_A	Q07075	0.944	-17.694	1.121
3	4ht_A	P15144	0.846	-22.211	1.068
4	6ryf_A	Q9NZ08	0.848	-19.026	1.038
5	7p7p_B	Q6P179	0.812	-22.099	1.033
.....					
22	7e7e_A	Q8BYF1	0.530	-18.480	0.715
23	Q9H4A4_F1_d1	Q9H4A4	0.508	-20.503	0.713
24	Q8IVL8_F1_d1	Q8IVL8	0.526	-17.927	0.705
25	P28838_F1_d1	P28838	0.515	-18.525	0.700

Download

- entire screen result [DOWNLOAD]

Link to download full prediction results

Figure 2. Exemplary submission and report pages of GalaxySagittarius-AF web server

GalaxySagittarius has been available since 2020 and an initial version of GalaxySagittarius-AF was introduced in 2022. Together, these two webserver have been utilized more than two thousand times (2,849 on 2023-01-09) across 49 nations. As of September of 2023, GalaxySagittarius-AF is considered as the primary webserver for protein target prediction, while GalaxySagittarius is no longer under active maintenance. GalaxySagittarius-AF is a part of the GalaxyWEB¹⁸ and is freely available at http://galaxy.seoklab.org/sagittarius_af without the need for registration.

4. Conclusion

Finding the protein targets that a given compound interacts with is crucial for evaluating off-target effects in drug development or understanding the broad activity of bioactive substances. GalaxySagittarius-AF covers the largest human protein space among available computational target screening methods for a given drug-like compound. Its structure-based target search method maximizes the utility of the expanded structural information, setting it apart from other target prediction servers that rely on ligand-based prediction methods. Additionally, the web server has undergone enhancements, operating two to three times faster than its predecessor. The updated report page provides comprehensive information on the sequence and structure of the predicted protein targets for further analysis.

5. References

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